

Project summary

SIH registration

Problem Statement ID: 26192

Problem statement: Flash Flood Prediction System for Hilly Regions using Multi-Source Data Theme

Project/app name: OraMet

Registered team name / ID / institute / members: FILL IN FROM SIH PORTAL

Copy-ready summary

OraMet is an Android-first prototype for local situational awareness during flood emergencies in hilly communities. It combines device-location context, current weather estimates from Open-Meteo when available, nearby mapped places from OpenStreetMap, emergency guidance, and a user-reviewed SMS draft addressed to India 112. It shows missing and approximate data explicitly and does not present unvalidated predictions, shelters, or directions as safe.

Purpose

The design goal is to bring available context and manual emergency actions into one resident-facing application while making source quality and uncertainty visible.

Implemented prototype

Available flows

React Native Android-first app, local navigation, location permission flow, approximate area selection, and local persistence.

Open-Meteo current-weather lookup when the public endpoint is reachable. Threshold labels are app heuristics only, not an IMD warning, gauge reading, or flood forecast.

OpenStreetMap/Overpass nearby mapped places with straight-line distance. Listings are not authority-verified, safe, or confirmed open.

Generic map directions with a warning that they are not hazard-aware and can encounter road closures or floodwater.

Emergency call action and prefilled SMS composer. The user must review and tap Send; delivery is not confirmed.

Explicitly not implemented

Authorized IMD/CWC/GSI/NHAI/ISRO/MOSDAC/WMO hazard feeds, validated flood lead time, or official shelter/route verification.

Automatic accelerometer shake detection, audible rescue siren, background push, SMS delivery receipt, and 112/CAD dispatch.

Production identity provider, hosted backend, responder dashboard, and emergency-service agreements.

Architecture and limitations

Current data path

React Native screens -> Zustand state -> device location and Open-Meteo public weather lookup; nearby mapped places use OpenStreetMap/Overpass; SQLite/AsyncStorage provide local app persistence.

Approximate selected areas are labeled and withheld from local risk scoring, nearby-place search, and location sharing. Missing live responses display unavailable instead of fabricated safe values.

Required for operations

Agency-approved and validated IMD/CWC/GSI feeds, authorized road-closure data, and maintained authority-verified shelter locations/status.

Hosted secure backend, identity, monitoring, privacy/security review, and compliant notification/communications provider.

Emergency-response agreements, field validation, accessibility testing, operational maintenance, and Android device/release testing.

Demo and validation

Suggested 90-second demo

Open the app and state that it is a prototype, not an operational warning service.

Show approximate area selection and explain that precise device location is required for nearby-place search/sharing.

Show live weather only if a valid response exists; otherwise show unavailable.

Show mapped OpenStreetMap places with the unverified notice and generic directions warning.

Show the missing official-feed state, then open and cancel an SMS draft. Do not claim dispatch or delivery.

Validation evidence

TypeScript check passed; lint passed; git diff whitespace check passed; 8 automated tests passed; Android Metro JavaScript bundle passed.

Native Android Gradle build was not run because the build environment has no Java/JDK. No release APK or device test is claimed.

Submission hygiene

Fill all registered team details from the SIH portal. Use actual screenshots/video from the running app.

Check the exact PDF copies before upload; cached exports created before the safety corrections may contain outdated claims.